

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

FACULTY : Dr. D. SUDHAGAR

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I **G . N a v e e n K u m a r R e d d y** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Naveen Kumar Reddy

QUESTIONS:05

Total Marks:25

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Ans:

A high initial learning rate allows the model to move quickly toward the global minimum in a convex loss function, avoiding slow progress at the beginning. Gradually decreasing (decay) the learning rate helps in fine-tuning near the minimum, reducing oscillations and ensuring convergence.

When it outperforms constant learning rate:

- When the loss surface is smooth and convex
- When starting far from the optimum
- When fast initial progress + precise final convergence is needed

Risks:

- Too high learning rate → overshooting or divergence
- Instability in updates
- May skip optimal solution

Conclusion:

This strategy balances speed (exploration) and accuracy (exploitation), making it more efficient than a constant rate in many cases.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

Ans:

Given Gaussian data with known variance:

MLE of mean (μ):

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

Effect of Outliers:

- Mean is highly sensitive → outliers can significantly shift μ
- Indicates MLE is not robust

Effect of Increasing Sample Size:

- Estimate becomes more accurate
- Variance of estimate decreases
- Converges to true mean (Law of Large Numbers)

Implication:

MLE is efficient but not robust, especially in presence of outliers.

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

Ans:

Challenges (10,000 features, 500 samples):

- Curse of dimensionality
- Overfitting
- High computational cost
- Sparse data representation

Proposed Pipeline:

1. **Feature Selection**
 - Remove irrelevant features (e.g., correlation, chi-square)
2. **Dimensionality Reduction**
 - Apply PCA to reduce features
3. **Regularization**
 - Use L1 (Lasso) or L2 (Ridge)
4. **Model Choice**
 - Use simple models (Logistic Regression, SVM)
5. **Cross-validation**
 - Ensure generalization

How it reduces overfitting:

- Reduces feature space
- Penalizes complexity
- Improves bias-variance trade off

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

Ans:

Why more hidden neurons increase power:

- Each neuron learns different features
- More neurons → more complex functions can be approximated
- Based on Universal Approximation Theorem

Why performance may not improve:

- Overfitting (memorizing training data)
- Increased computational cost
- Vanishing gradients (with sigmoid)
- Poor generalization

Conclusion:

More neurons increase capacity, but optimal size is required for best performance.

5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

Ans:

Impact:

1. Gradient Updates:

- Gradients become sparse or unstable
- Hard to find meaningful direction

2. Convergence Speed:

- Slower convergence
- Requires more iterations

3. Generalization:

- High risk of overfitting
- Poor performance on unseen data

Techniques to Overcome:

1. **Dimensionality Reduction (PCA, Autoencoders)**
 - Reduces feature space → better gradients
2. **Regularization (Dropout, L2)**
 - Prevents overfitting
3. **Batch Normalization**
 - Stabilizes gradients
4. **Feature Selection**
 - Removes irrelevant features

Conclusion:

Reducing dimensionality and controlling model complexity improves training efficiency and generalization.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

